

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“Global Logistics-Celonis”

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Niharika Rajendra Ankalkoti - 1DS21AI036

Conducted at



Anheuser-Busch InBev

Department of Artificial Intelligence and Machine Learning

Dayananda Sagar College of Engineering Bangalore-78

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CERTIFICATE

This is to certify that the Internship titled “**Global Logistics-Celonis**” carried out by **Ms NIHARIKA RAJENDRA ANKALKOTI [1DS21AI036]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering, in Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (20AI8ICINT)

**Signature of Reporting
Head**
Devesh Jindal
Manager
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**Signature of Internship
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Name of the Examiner

Signature with Date

1) _____

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DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

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DECLARATION

I, **Niharika Rajendra Ankalkoti**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **Anheuser Busch InBev**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2023- 2024.

Date:12/04/25

USN: 1DS21AI036

Place: Bangalore

NAME: Niharika R A

OFFER LETTER



17/02/2025
Niharika R A

Dear Niharika,

Subject: Internship Letter

Congratulations!

We Dream Big to Create a Future with More Cheers!

We at AB InBev GCC Services India Private Limited ("GCC" or the "Company") are delighted to extend our offer to you as part of our internship program based in Bangalore.

AB InBev GCC Services India Private Limited is a group/associate company of Anheuser Busch InBev ("AB InBev") committed to driving growth that leads to better living for more people in more places. We take the development of our people seriously and help you dream bigger than what you think is possible. This is an exciting opportunity to intern with AB InBev and to challenge yourself with innovative and impactful projects. AB InBev is a leading global brewer and one of the world's top five consumer product companies.

Your Internship with AB InBev GCC Services India Private Limited shall commence from 24/02/2025 to 30/06/2025 subject to your acceptance of these Terms and Conditions. Please sign and return a copy of this letter and the Terms and Conditions to indicate your acceptance of this Tech Internship. The Terms and Conditions of your Internship are included below as Schedule A.

You will be given an extensive orientation on AB InBev upon commencement of the Internship.

We look forward to working with you and would like to take this opportunity to wish you every success during your Internship.

Yours sincerely,
Abhijeet Kadam
Lead – Talent Acquisition

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Mr. Devesh Jindal**, Manager, ABInbev for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, **Prof. Kavya D N**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

NIHARIKA R A [1DS21AI036]

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CHAPTER 1

INTRODUCTION

In today's data-driven enterprise landscape, real-time decision-making and operational transparency have become critical components for organizational success. One of the key enablers of this transformation is **KPI-driven process automation**, which helps enterprises monitor performance, detect inefficiencies, and streamline workflows. During my internship at the **Global Command Center (ABI)**, I had the opportunity to work at the intersection of **process mining, data engineering, and business intelligence**, where I was exposed to industry-grade tools such as **Celonis, Azure Data Factory (ADF), Power BI, and SharePoint-integrated data sources**.

The internship was structured around creating a **robust, automated KPI consumption pipeline** that ingests performance metrics from multiple sources—manual files, Celonis platforms, and Power BI dashboards—into a centralized **KPI Data Lake**. My responsibilities involved setting up data extraction logic using Python, configuring ADF pipelines to transform and load data, and building Celonis dashboards to analyze the performance of logistics processes globally. This end-to-end flow helped ensure that all KPI data was accurately represented and accessible for analytical decision-making.

Another major learning aspect of the internship was working with **Celonis Studio and ML Workbench**, where I explored the capabilities of **Process Query Language (PQL)** to filter, join, and aggregate business process data. These analyses were essential for identifying manual interventions in logistics processes like Order Management and Inventory Deployment, and for promoting **touchless automation**—a methodology aimed at reducing human dependency and boosting efficiency.

The internship provided a holistic view of how organizations utilize **process intelligence and automation** to improve their operational performance. It also deepened my understanding of **data orchestration, pipeline engineering, and KPI governance**, which are increasingly vital in fields like Artificial Intelligence and Machine Learning. Through hands-on experience, I gained practical knowledge in integrating and aligning complex data sources, enabling me to contribute meaningfully to enterprise-scale data solutions.

This report outlines my entire internship journey, detailing the scope of work, tools used, challenges faced, solutions implemented, and key learnings acquired during my tenure with the ABI Global Command Center.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

ANHUESER BUSCH INBEV



Introduction

Anheuser-Busch InBev SA/NV, commonly known as **AB InBev**, is the world's largest brewer and a multinational drink and brewing company headquartered in **Leuven, Belgium**. As of 2023, the company ranks **72nd in the Forbes Global 2000** and operates with a presence in over **150 countries** through a massive portfolio of around **630 beer brands**, including some of the most iconic names like **Budweiser, Corona, Stella Artois**, and **Beck's**.

AB InBev has global offices in **New York City**, with major regional headquarters in **São Paulo, London, St. Louis, Mexico City, Bremen, Johannesburg**, and other strategic locations. The company is publicly listed with its primary listing on the **Euronext Brussels**, and secondary listings on stock exchanges in **Mexico City, Johannesburg**, and **New York (NYSE: BUD)**.

Historical Background

The roots of AB InBev trace back to the **Interbrew** group in Belgium, which itself was formed from the merger of two historic breweries – **Artois** (established in 1366) and **Piedboeuf** (1812). Interbrew expanded aggressively during the 1990s and 2000s by acquiring breweries in **Canada (Labatt)**, **Russia (Sun)**, **UK (Bass and Whitbread)**, and **Germany (Beck's)**. In 2004, **Interbrew** merged with **AmBev**, a leading Brazilian beverage company formed through the merger of **Brahma** and **Antarctica**, creating **InBev**.

AmBev, based in São Paulo, is Latin America's largest beverage company and remains a significant AB InBev subsidiary, known for its beer and soft drink brands, including **Guaraná Antarctica** and **Sukita**. InBev became the largest brewing company globally by 2004, continuing to grow by acquiring competitors and strengthening its market dominance.

In 2008, InBev made its landmark move by acquiring the American brewing giant **Anheuser-Busch** for **\$52 billion**, forming the company we now know as **Anheuser-Busch InBev (AB InBev)**. Anheuser-Busch, founded in 1852 in St. Louis, Missouri, was the largest brewer in the United States and the maker of Budweiser, Michelob, and Natural Light.

Major Mergers and Global Expansion

A defining moment in AB InBev's growth came in **2016** when it acquired **SABMiller**, the second-largest brewer at the time, for **\$107 billion**. This acquisition further cemented AB InBev's dominance, giving it a **28% global market share** in beer sales and making it one of the top consumer goods companies in the world.

To satisfy anti-trust conditions, AB InBev divested key SABMiller assets including the **Miller** brand portfolio (sold to **Molson Coors**) and the **Peroni, Grolsch, and Meantime** brands (sold to **Asahi**). Additionally, it sold stakes in its operations across **Central and Eastern Europe** to Asahi Breweries. Some other notable acquisitions and divestments include:

- **Grupo Modelo (Mexico)** in 2013 for \$20.1 billion, known for **Corona**
- Reacquisition of **Oriental Brewery (South Korea)** in 2014
- Establishment of **Bud Analytics Lab** in 2013 to focus on AI and data analytics
- Partnership with **Tilray** in 2018 to explore cannabis-infused beverages
- Sale of various craft beer and energy drink brands to Tilray in 2023

Business Model and Strategic Focus

While beer is AB InBev's core product, the company is increasingly leveraging **data science, analytics, and AI** to enhance its operations. Through its **Global Command Centers (GCCs)** and analytics labs, it focuses on supply chain optimization, predictive maintenance, digital transformation, and process automation. These centers play a critical role in monitoring logistics, reducing manual interventions, and enabling **touchless business operations**.

AB InBev's mission is to "**Dream Big to Create a Future with More Cheers.**" The company's values emphasize ownership, innovation, consumer focus, and sustainability. With initiatives like the **2025 Sustainability Goals**, AB InBev is working toward eco-friendly practices including **carbon neutrality, recyclable packaging, and water conservation**.

Recent Developments

In 2023, AB InBev initiated a share buyback program valued up to **\$1 billion**, showing confidence in its financial stability despite its significant liabilities (US\$95.5 billion in 2019). The company continues to make strategic shifts toward premiumization, non-alcoholic innovations, and digital expansion.

Despite challenges from global economic fluctuations, changing consumer habits, and increasing competition, **AB InBev remains a global powerhouse**, driven by its vast brand portfolio, operational scale, and commitment to leveraging technology for better business outcomes.

ABOUT THE COMPANY

About Global Command Center

Global Command Center (ABI) is a centralized data monitoring and analytics hub that operates as part of a global enterprise ecosystem. The organization plays a pivotal role in driving operational efficiency by enabling real-time visibility into key business metrics, especially those related to **logistics performance, supply chain movement, and touchless automation**. With a data-first approach, the center serves as the backbone for integrating diverse data sources and converting raw information into meaningful Key Performance Indicators (KPIs) that guide strategic decision-making across business units.

The **Global Command Center** works closely with various teams to create **automated and intelligent dashboards**, analyze process bottlenecks, and identify manual interventions that disrupt efficiency. A major focus is on promoting **touchless practices** in logistics—an initiative that involves minimizing human input through intelligent automation of repetitive tasks, predictive analysis, and standardized reporting mechanisms.

To support these objectives, the organization utilizes cutting-edge tools such as **Celonis** for process mining, **Azure Data Factory (ADF)** for pipeline orchestration, **Power BI** for data visualization, and **SharePoint** for storing and accessing structured manual files. These tools work in harmony to ensure seamless ingestion, transformation, and analysis of KPI data across multiple regions and departments.

One of the distinguishing features of the Global Command Center is its ability to **aggregate data from multiple environments**—including manual file systems, enterprise data lakes, and third-party applications—into a consolidated format used by leadership teams for performance evaluation. It also leverages **Celonis ML Workbench** to integrate Python scripts for advanced analytics, and builds scorecards that measure zone-wise efficiency in operations such as **Order Management** and **Inventory Deployment**.

By aligning technology with business intelligence, the organization ensures standardized monitoring, faster anomaly detection, and quicker decision loops across global functions. Through its structured approach to data analysis and automation, the Global Command Center exemplifies how modern enterprises are evolving toward **data-powered excellence**.

CHAPTER 3

SCOPE OF WORK

During my internship at **AB InBev's Global Command Center (GCC)**, I was actively involved in multiple critical data engineering and process automation projects focused on **KPI data ingestion, transformation, and dashboard creation**. The primary goal was to establish an end-to-end automated pipeline that enabled seamless flow of KPI data from various internal sources into a centralized **KPI Data Lake**.

My responsibilities and contributions were distributed across three main key deliverables:

1. Integration of Manual KPI Files

- I worked on automating the data pipeline to extract KPI metrics from **manual Excel files** stored on **SharePoint**.
- Using **Azure Data Factory (ADF)**, I developed workflows that:
 - Extracted data using **Python scripts** hosted on **ADB (Azure Databricks)**.
 - Transformed and validated the data for consistency.
 - Loaded the cleaned data into the **KPI Data Lake**, which served as the central storage used by product teams for KPI access.

2. KPI Data Integration from Celonis

- I collaborated with the **Celonis Studio and ML Workbench** tools to retrieve processed KPI data from various **process mining dashboards**.
- Wrote **PQL (Process Query Language)** queries to filter, join, and aggregate business logic for logistics-related KPIs.
- Integrated the output tables into the pipeline and automated their ingestion into the **KPI Data Lake**, ensuring real-time availability to analytics teams.

3. Power BI Data Source Automation

- Contributed to automating the KPI data sourced from **Power BI dashboards** and embedded Excel datasets.
- Similar to the manual file flow, data was:
 - Extracted using Python-based extraction logic.
 - Processed through ADF pipelines and stored in the data lake in a standardized format.

- Ensured the Power BI dashboards aligned with updated KPI definitions and helped sync them with the centralized reporting system.

Cross-functional Contributions

- Developed **Celonis dashboards and automation scorecards** to track **Order Management** and **Inventory Deployment** KPIs across global zones.
- Supported the **touchless logistics initiative** by identifying process inefficiencies and manual interventions in real-time.
- Attended review sessions with cross-functional teams to validate data accuracy and KPI alignment.

CHAPTER 4

LEARNING OBJECTIVES

During my internship at AB InBev's Global Command Center, I gained hands-on experience across the full data lifecycle—from sourcing and orchestrating data, to visualizing it for business insights. The internship not only strengthened my technical foundation but also enhanced my understanding of enterprise-grade automation and performance tracking systems. Below are the core learning areas and detailed insights:

1. Understanding the KPI Lifecycle: Source → Pipeline → Visualization

I developed a strong grasp of how Key Performance Indicators (KPIs) are generated, tracked, and consumed within a global organization. The KPI lifecycle involves:

- **Source:** Data originates from multiple platforms such as Excel files on SharePoint, Celonis dashboards, or Power BI exports. These sources contain raw performance metrics related to logistics, procurement, inventory, etc.
- **Pipeline:** Data is extracted, cleaned, transformed, and loaded (ETL process) through tools like Azure Data Factory (ADF). I helped configure pipelines that automated these tasks, ensuring data flowed seamlessly from source to destination.
- **Visualization:** Once processed, the data is stored in the KPI Data Lake, and used to feed dashboards built in Celonis and Power BI, allowing business users to make informed decisions using real-time performance insights.

This end-to-end lifecycle helped me understand the importance of data standardization, timely ingestion, and centralized access in enabling consistent KPI reporting across geographies.

2. Learning Celonis Tools: Studio, Analyses, and PQL Scripting

Celonis is a powerful process mining platform that reveals bottlenecks and inefficiencies in business operations. I worked with:

- **Celonis Studio** to design visual dashboards that mapped logistics workflows like Order Management and Inventory Deployment.
- **Celonis Analyses** to create and modify pre-built visual reports and views.
- **PQL (Process Query Language)**, a specialized SQL-like language used to filter, aggregate, and join business data within Celonis. I wrote PQL queries to define custom KPIs and track user interactions across the order lifecycle.

Through these tools, I contributed to automation scorecards that measured manual intervention rates and touchless processing percentages—essential metrics for driving AB InBev’s digital transformation.

3. Working with Azure Data Factory (ADF) for Data Orchestration

One of the critical tools I used was Azure Data Factory, a cloud-based data integration service. My tasks included:

- Creating and managing pipelines to automate the ETL process.
- Scheduling trigger-based jobs that executed Python scripts on Azure Databricks (ADB).
- Monitoring pipeline runs, identifying failures, and performing debug analysis.

ADF helped integrate data from sources like SharePoint and Celonis into a unified data lake. The platform’s low-code interface made orchestration scalable and efficient, giving me real-world experience in cloud-based data engineering.

4. Gaining Exposure to BI Tools: Power BI and Excel Integrations

I interacted extensively with Power BI—a business intelligence platform used to visualize KPIs and build dashboards for leadership and product teams. My exposure included:

- Syncing Power BI reports with the KPI Data Lake.
- Integrating Excel-based KPI files with Power BI for dynamic refreshes.
- Standardizing data definitions across Power BI and Celonis to avoid inconsistencies.

I also worked with Excel as a KPI source format, writing scripts to ingest data and validate file structures before transformation. These tasks emphasized the importance of data governance and version control in real-world reporting environments.

5. Analyzing Logistics Processes Using Touchless Automation Frameworks

AB InBev’s Touchless Automation Framework is a methodology that aims to reduce human intervention in logistics processes by promoting automation, standardization, and real-time tracking. My contributions involved:

- Building scorecards that assessed the level of manual intervention in tasks like order creation, approval, and shipment.
- Identifying bottlenecks using Celonis process flows and filters.
- Benchmarking automation maturity across global zones and suggesting areas for improvement.

This experience helped me understand how enterprises leverage process intelligence and automation to improve productivity, ensure compliance, and enhance customer satisfaction.

CHAPTER 5

CHALLENGES AND SOLUTION

During my internship, I encountered several technical and process-related challenges while working on KPI integration, automation flows, and dashboard development. These challenges provided valuable learning opportunities and exposed me to real-world enterprise-level problem-solving. Below is a detailed account of the major challenges I faced and the corresponding solutions I implemented:

Challenge 1: Integrating Heterogeneous Data Sources

AB InBev's KPI data originates from multiple systems including manual Excel files, Celonis process mining tools, and Power BI dashboards. These sources differ in structure, format, frequency of updates, and storage methods (e.g., SharePoint vs. ADB).

Solution:

To standardize this fragmented ecosystem:

- I built and tested ADF (Azure Data Factory) pipelines to automate extraction and transformation of KPI files from various sources.
- Integrated these pipelines with Azure Blob Storage, allowing a common staging area before loading data into the KPI Data Lake.
- Ensured that output schema across sources matched the data lake requirements, enabling seamless downstream analysis.

Challenge 2: Writing Efficient PQL Queries for Dynamic KPI Filtering

Celonis PQL (Process Query Language) is used to create logic-driven KPI dashboards. The challenge was writing queries that could dynamically adjust based on filters such as region, time, or process status without affecting performance or dashboard responsiveness.

Solution:

- I learned the advanced capabilities of PQL including PU-functions, aggregation techniques, and context filters.
- Used joins and subqueries to dynamically link related data models (e.g., order lifecycle stages).
- Created modular queries that could adapt to different dashboard states, improving performance and scalability.

Challenge 3: Creating Automation Flows with Minimal Manual Intervention

Manual intervention in data refreshes, error handling, or updates defeated the purpose of "touchless" KPI tracking. Designing an end-to-end automated flow that ran error-free and required zero human intervention was essential but complex.

Solution:

- Designed ADF workflows with built-in triggers, error logging, and conditional paths that allowed retries in case of failures.
- Integrated Python scripts hosted in Azure Databricks (ADB) for dynamic data extraction from Excel files.
- Scheduled and monitored jobs to ensure data was pushed to the KPI Data Lake on a predefined cadence, with email alerts in case of job failures.

Challenge 4: Handling Data Discrepancies Across Versions

Multiple teams worked on different KPI sources, which often led to discrepancies in how KPIs were defined, calculated, or labeled. This caused confusion and inconsistencies across dashboards.

Solution:

- Worked with business and data stakeholders to establish a single source of truth for each KPI.
- Used Celonis filters and validation logic to cross-check discrepancies across regions and sources.
- Built standardized dashboards that aligned with official KPI definitions and visualized the flow of each metric from origin to outcome.

HARDWARE AND SOFTWARE REQUIREMENTS

Software requirements:

- Operating System: Windows 10/11
- Celonis EMS: Process mining
- Azure Data Factory: Data orchestration
- Azure Databricks: Python scripting
- Python: Automation scripts
- Power BI: KPI visualization
- Excel: Source data
- SharePoint: File storage

Hardware Requirements:

- Processor: Intel i5/i7
- RAM: 8–16 GB
- Storage: 256 GB SSD
- Display: HD Monitor
- Input Devices: Keyboard, Mouse
- Internet: High-speed connection
- Cloud Access: Azure services

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

Throughout my internship at AB InBev's Global Command Center, I actively contributed to the development and optimization of KPI tracking systems, process automation workflows, and business dashboards that supported the organization's global logistics and operations. My work spanned across multiple platforms and tools including Celonis, Azure Data Factory, Azure Databricks, SharePoint, and Power BI. Below are the key achievements and my detailed contributions:

1. Built PQL Dashboards in Celonis for Real-Time KPI Tracking

One of my core responsibilities was designing and implementing Celonis dashboards that visualized KPIs in real-time. Using PQL (Process Query Language), I created queries that filtered, grouped, and aggregated logistics data such as order lifecycle statuses, delivery timelines, and process efficiency rates.

- Developed custom metrics to monitor manual vs. automated processes.
- Implemented filters to allow dynamic views based on region, process stage, or timeframe.
- Enabled leadership and process owners to gain live insights into operational bottlenecks and touchless automation rates.

This not only enhanced visibility into logistics processes but also helped drive data-based decision-making across global teams.

2. Designed Data Pipelines Using ADF and ADB to Automate Data Flow

I was involved in building automated data pipelines using Azure Data Factory (ADF) and Azure Databricks (ADB) to eliminate manual data processing steps.

- Wrote Python scripts in Databricks to extract and clean data from Excel files stored in SharePoint.
- Used ADF to orchestrate end-to-end ETL workflows that loaded cleaned KPI data into the Azure Blob Storage and then into the KPI Data Lake.
- Configured pipeline triggers and alerts for automated execution, reducing delays and human errors.

These pipelines served as the backbone for feeding accurate, up-to-date KPI data into reporting tools like Power BI and Celonis.

3. Performed KPI Integration from SharePoint, Power BI, and Celonis

A major achievement was successfully integrating KPI data from multiple sources and standardizing it for consistent use across the enterprise.

- Extracted KPI data from manual Excel files stored on SharePoint using custom Python logic.
- Mapped and aligned KPI definitions from Power BI dashboards and Celonis to ensure accuracy and uniformity.
- Ensured that the KPI Data Lake served as a single source of truth for all reporting layers.

This integration work played a critical role in building a reliable and trustworthy KPI tracking ecosystem used by business and product teams globally.

4. Created Views and Scorecards for Order Management and Inventory Deployment

I contributed to the development of Celonis views and automation scorecards that visualized key metrics related to logistics performance, especially in:

- Order Management: Tracked the end-to-end order cycle including creation, approval, fulfillment, and delivery.
- Inventory Deployment: Monitored stock movements, deployment efficiency, and delays across different zones.

These views were customized to show:

- Zone-wise performance comparisons.
- Percentage of touchless processing.
- Manual interventions and their root causes.

They helped highlight areas of inefficiency and prompted targeted process improvements.

5. Contributed to Touchless Automation Practices Across Zones Globally

AB InBev promotes touchless automation, which refers to reducing manual intervention in key business processes. I played a supporting role in this transformation initiative by:

- Identifying high-touch processes using Celonis process mining.
- Highlighting manual activities and exceptions through real-time dashboards.
- Building KPIs that measured automation maturity across different regions.

My work directly supported the company's goal of implementing end-to-end digital workflows in logistics and supply chain management, and contributed to benchmarking automation adoption globally.

CHAPTER 7

REFLECTION AND ANALYSIS

My internship at **AB InBev's Global Command Center (GCC)** was a transformative learning experience that exposed me to real-world enterprise-level data ecosystems. Over the course of the internship, I gained hands-on experience with end-to-end data pipeline development, business process mining, KPI reporting, and touchless automation practices. This chapter outlines my key reflections and the professional growth I experienced during the internship.

1. Exposure to Enterprise Data Workflows

One of the most valuable aspects of the internship was understanding how data flows through an organization—from fragmented, raw files stored in SharePoint to well-structured KPI dashboards consumed by global teams. I observed the importance of designing scalable, error-tolerant, and automated pipelines that could ingest, transform, and store data with minimal human input. Tools like **Azure Data Factory** and **Azure Databricks** gave me a practical understanding of **cloud-based data orchestration**, a highly relevant skill in today's digital-first businesses.

2. Learning the Power of Process Mining with Celonis

Working with **Celonis EMS** was an eye-opening experience. I learned how companies visualize and analyze their end-to-end business processes, identifying bottlenecks, inefficiencies, and opportunities for automation. The ability to use **Process Query Language (PQL)** to derive meaningful insights from massive datasets taught me how **data-driven decision-making** is embedded in modern business intelligence strategies. I developed PQL queries for KPI calculations, created custom filters, and helped build real-time dashboards used by stakeholders across different regions.

3. Development of Technical and Analytical Skills

This internship helped me significantly strengthen my technical skillset. I learned:

- How to write clean and efficient Python scripts for data extraction and validation.
- How to manage **ETL pipelines** using ADF with proper error handling and triggers.
- How to visualize KPIs effectively using **Power BI** and Celonis. Additionally, I practiced **version control with Git**, used collaborative platforms like Microsoft Teams, and learned how to document technical workflows clearly for future reference.

Beyond tools, the experience taught me **data integrity**, **KPI governance**, and the importance of aligning technical solutions with business requirements.

4. Real-World Problem Solving and Collaboration

One of the biggest learning takeaways was dealing with **real-world data challenges**—such as inconsistent formats, evolving KPI definitions, and the integration of data from diverse sources. I often had to collaborate with multiple teams to clarify business logic, resolve data mismatches, and ensure accuracy in dashboard outputs. These experiences taught me the importance of **clear communication, stakeholder alignment, and agile problem-solving** in a fast-paced enterprise environment.

5. Understanding the Impact of My Work

What made this internship truly rewarding was realizing that the work I was doing—whether building a pipeline or fixing a dashboard—had a direct impact on global business operations. The dashboards I helped build enabled regional teams to monitor performance, detect inefficiencies, and optimize logistics. My contributions supported the company's larger goal of **digital transformation through touchless automation**—a movement toward intelligent, zero-touch processes powered by technology.

CHAPTER 8

RECOMMENDATIONS

During my internship at AB InBev's Global Command Center, I had the opportunity to engage in several impactful projects that deepened my understanding of process mining, data engineering, and automation. While the internship was highly enriching, I believe there are areas where the experience could be made even more beneficial for future interns. The following are my recommendations based on my observations and learnings.

To begin with, I would recommend offering deeper training in process mining tools, especially Celonis. While I gained hands-on experience with Celonis Studio and PQL (Process Query Language), a more structured onboarding session could have accelerated my learning. Introducing a dedicated bootcamp or workshop in the initial phase of the internship would allow interns to become familiar with the capabilities of process mining tools in a guided manner. Access to sample dashboards, practice datasets, and curated exercises would further enable interns to explore use cases independently and build a stronger foundation in analyzing process efficiencies and bottlenecks.

Another useful enhancement would be the creation of standard documentation templates for pipeline design. Since a large portion of the internship involves working with Azure Data Factory and integrating various data sources, it would be highly beneficial to provide templates that document the input, transformation, and output flow of data pipelines. This documentation could include details on naming conventions, data lineage, scheduling triggers, and error handling logic. Well-structured templates not only aid in building organized workflows but also support easier handovers and better collaboration among teams.

Additionally, allowing cross-functional exposure across tools and departments would add significant value to the learning experience. Interns are typically assigned to work within a single tool such as Celonis, Power BI, or ADF, which is great for specialization. However, even limited exposure to adjacent tools can offer valuable context and a more holistic view of the data ecosystem. For example, interns working on Celonis dashboards could benefit from understanding how data is transformed upstream in ADF or visualized further downstream in Power BI. Organizing short knowledge-sharing sessions, team demos, or even rotational mini-projects can help interns gain a broader perspective of interconnected workflows.

Lastly, setting up demo environments for experimentation would be highly effective. Since most interns work with real production datasets, opportunities for trial-and-error are limited. Having access to sandbox environments—with simulated KPI data, mock dashboards, and test pipelines—would allow interns to experiment more freely, explore different design approaches, and learn from their mistakes without affecting live operations.

CHAPTER 9

CONCLUSION

My internship at **AB InBev's Global Command Center** has been a profoundly enriching and transformative experience, providing me with hands-on exposure to enterprise-level technologies and real-world business process automation. Through this journey, I gained a deeper understanding of how large-scale organizations manage, monitor, and optimize their operations using data-driven strategies. The experience bridged the gap between theoretical learning and practical application, offering me a glimpse into the inner workings of global digital ecosystems.

One of the most valuable aspects of this internship was learning the significance of **data automation and orchestration** in driving operational efficiency. Working with **Azure Data Factory (ADF)** allowed me to create automated pipelines that replaced manual, repetitive tasks with seamless and scalable data flows. Through real-time monitoring and trigger-based execution, I saw how automated systems could enhance accuracy, reduce delays, and empower teams to focus on insights rather than data collection.

The use of **Celonis** for process mining was another standout learning. By working with **PQL queries** and dashboard creation, I learned how business processes could be visualized and analyzed at a granular level. This helped identify inefficiencies and manual interventions, leading to data-backed recommendations for touchless automation. I realized how process intelligence plays a crucial role in identifying root causes and improving throughput across various operational functions, particularly in areas like Order Management and Inventory Deployment.

In addition, building **interactive dashboards using Power BI** introduced me to the power of data visualization in storytelling. I learned to transform raw metrics into meaningful performance indicators that were easily interpretable by decision-makers across regions. These visualizations not only enabled faster insights but also ensured alignment with AB InBev's strategic goals.

More broadly, the internship taught me the importance of **collaboration, documentation, and standardization**. I engaged with global teams, wrote code with clarity, participated in review sessions, and adhered to best practices in version control and project tracking. These soft skills are as essential as technical know-how and will undoubtedly support me in future roles.

Overall, this internship has equipped me with a practical toolkit of skills in **data engineering**,

process analytics, and BI integration. More importantly, it has boosted my confidence to take on real-world challenges involving AI/ML-driven process optimization and enterprise automation. I now look forward to applying these learnings in future projects and career opportunities where I can contribute to building intelligent, data-powered solutions that drive meaningful business outcomes.

CHAPTER 10

REFERENCES

1. Celonis Training: Build Analyses, Process Mining, Studio Dashboards

The official Celonis Training Portal was a foundational resource in helping me understand how to create and configure process mining dashboards. I referred to modules related to Celonis Studio, where I learned how to design analysis sheets, use filters, and set up custom KPI visualizations. The training also covered best practices in process mapping, conformance checking, and touchless automation metrics, all of which I applied while working on logistics dashboards during my internship.

2. Azure Data Factory Documentation

Microsoft's Azure Data Factory (ADF) documentation served as a go-to reference for building, configuring, and debugging data pipelines. I explored topics such as Linked Services, Datasets, Activities, and Trigger Scheduling. The documentation also provided guidance on integrating ADF with Azure Databricks, handling parameterization, and implementing data flow monitoring—all of which were directly relevant to the automated workflows I designed for ingesting KPI data from SharePoint and Excel sources.

3. Power BI Official Docs

The Power BI Documentation provided by Microsoft helped me understand how to build and modify visual reports and dashboards. I used it to explore data modeling, measure creation, visual customization, and dashboard publishing techniques. Additionally, it supported my work in aligning Power BI dashboards with data pulled from the KPI Data Lake, ensuring consistency in performance reporting across tools.

4. PQL (Process Query Language) Overview – Celonis

Understanding PQL, the query language native to Celonis, was critical for writing scripts that powered custom KPIs and logic-based filters. I referred to the PQL language reference guide to learn about functions such as `PU_SUM()`, `FILTER()`, `JOIN()`, and nested expressions. These functions helped me retrieve meaningful insights from process data, apply logic for calculating automation rates, and build condition-driven analysis sheets.

5. Internal Team Documentation – ABI Global Command Center

In addition to external documentation, I had access to internal team documentation maintained by the GCC team at AB InBev. These resources included KPI dictionaries, dashboard specifications, pipeline configuration notes, and deployment standards. This internal knowledge base was essential for ensuring consistency in KPI definitions, aligning with the expectations of stakeholders, and contributing effectively to ongoing projects without disrupting established processes.